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APPLICATION NUMBER:

208078Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

Office of Clinical Pharmacology Review Addendum (Erratum)

NDA Number	208078
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Submission Date	03/28/2018
Submission Type	505(b)(1) Original NME NDA
Brand Name	FIRDAPSE®
Generic Name	Amifampridine phosphate
Dosage Form/Strength and Dosing Regimen	Immediate Release Tablets 10 mg, scored on one side The recommended starting dose is 15-30 mg a day divided over three or four doses a day. Dose is not to exceed a maximum daily dose of 80 mg per day.
Route of Administration	Oral
Proposed Indication	(b) (4) treatment of Lambert-Eaton Myasthenic Syndrome (LEMS)
Applicant	Catalyst Pharmaceuticals, Inc.
Associated IND	IND 106263
OCP Review Team	Bilal AbuAsal, Ph.D.; Katarzyna Drozda, Ph.D.; Christian Grimstein, Ph.D.; Kevin Krudys, Ph.D.; Sreedharan Sabarinath, Ph.D.
OCP Final Signatory	Ramana Uppoor, Ph.D.

The purpose of addendum is to correct a typographical error with the submission date of the NDA listed in the cover page of the Clinical Pharmacology Filing Review (DARRTS dated 6/22/2018) and Final Clinical Pharmacology Review (DARRTS dated 09/14/2018).

The original reviews listed date of NDA submission as 03/28/2015 instead of 03/28/2018.

The correct date of submission is 3/28/2018.

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

BILAL S ABU ASAL
10/04/2018

SREEDHARAN N SABARINATH
10/04/2018

Office of Clinical Pharmacology Review

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1. EXECUTIVE SUMMARY

Catalyst Pharmaceuticals, Inc. submitted an original new drug application (NDA) seeking approval for amifampridine phosphate tablets (FIRDAPSE®) for the (b) (4) treatment of Lambert-Eaton Myasthenic Syndrome (LEMS), a rare and severely debilitating disease.

FIRDAPSE® was approved for this indication by the European Medicines Authority (EMA) and has been marketed in the European Union by BioMarin Pharmaceutical, Inc. since December 2009. In the United States, amifampridine has been available for the treatment of LEMS under a compassionate use program since the 1990s.

LEMS is an autoimmune disease characterized by muscle weakness and fatigability. These symptoms are caused by the impairment of acetylcholine (ACh) release at the peripheral neuromuscular junctions, causing a loss of neuromuscular transmission. As a voltage-dependent potassium channel blocker, amifampridine causes depolarization of the presynaptic membrane and slows down or inhibits repolarization. This prolonged depolarization results in the opening of slow voltage-dependent calcium channels and allows a subsequent influx of Ca^{2+} to increase ACh release and consequently, improves neuromuscular transmission.

In this NDA, the efficacy of FIRDAPSE® was established based on data from two placebo-controlled Phase III studies (Study LMS-002 and Study LMS-003). Sixty-four adult subjects with LEMS (age 21 to 88 years) were enrolled in these studies. Subjects were required to be on adequate and stable dose of amifampridine prior to entering the randomized discontinuation phase of both studies. The primary measures of efficacy were the change in the Quantitative Myasthenia Gravis (QMG) and Subject Global Impression (SGI) scores at the end of the discontinuation period of the study compared with baseline.

In addition, the applicant submitted five Phase I clinical pharmacology studies to support this NDA and to characterize the effect of intrinsic and extrinsic factors on the plasma exposure of FIRDAPSE® (i.e. food effect, renal impairment, acetylation genotype status and TQT studies)

The primary focus of this review is to evaluate:

1. Acceptability of the proposed starting dose and maximum daily dose, and
2. Need for dose adjustments based on intrinsic and extrinsic factors.

1.1 Recommendations

The Office of Clinical Pharmacology has reviewed the information submitted under NDA 208078 and recommends approval of amifampridine phosphate tablets (FIRDAPSE®) for the (b) (4) treatment of Lambert-Eaton Myasthenic Syndrome (LEMS) in adults.

Key review issues with specific recommendations and comments are summarized below:

Review Issue	Recommendations and Comments
Pivotal or supportive evidence of effectiveness	The efficacy of FIRDAPSE® was established based on data from two placebo-controlled Phase III studies (Study LMS-002 and Study LMS-003). The primary measures of efficacy were the change from baseline in the Quantitative Myasthenia Gravis (QMG) score and Subject Global Impression (SGI) score at the end of the discontinuation period.
General dosing instructions	• FIRDAPSE® is to be taken orally in divided doses 3 to 4 times per day. • The recommended starting dose should range from 15-30 mg a day. The lower starting dose of 15 mg should be considered when initiating treatment in subjects with renal or hepatic function impairment or who are known slow acetylators. • The dose can be escalated to a maximum of 80 mg per day by increasing the dose in 5 mg per day increments every 3 or 4 days. • The maximum unit dose allowed is 20 mg. • FIRDAPSE® can be administered without regard to food.
Dosing in patient subgroups (intrinsic and extrinsic factors)	Intrinsic factors like renal impairment and acetylation status result in increase in amifampridine exposure. The recommended dose titration based on tolerability and efficacy is expected to allow for appropriate individualization of dose for all patients. For the same reason, capping the maximum daily dose based on these factors is not required. There were LEMS patients who were slow acetylators stabilized on the recommended maximum daily dose of 80 mg in Phase III studies. However, the inclusion of a lower starting total daily dose of 15 mg is recommended to ensure slower dose titration in patients with these intrinsic factors.

	Amifampridine phosphate undergoes hepatic metabolism. However, the effect of hepatic impairment was not evaluated and the applicant is planning to evaluate it after approval. Finally, dose adjustment is not required based on demographic factors such as age, sex, race, ethnicity and body weight.
Bridge between the to-be-marketed and clinical trial formulations	The formulation of amifampridine phosphate used in the pivotal efficacy study is identical to the to-be-marketed formulation.

1.2 Post-Marketing Requirements and Commitments

Amifampridine is predominantly metabolized by N-acetyltransferase in the liver. There is no clinical experience with amifampridine in subjects with hepatic impairment. The applicant has committed to conduct a hepatic impairment study to evaluate the impact of hepatic function impairment on the exposure to amifampridine (Type B meeting minutes in DARRTS dated 12/20/2013).

2. SUMMARY OF CLINICAL PHARMACOLOGY ASSESSMENT

2.1 Pharmacology and Clinical Pharmacokinetics

Mechanism of Action: Amifampridine is a voltage-gated potassium (K⁺) channel blocker. The blockade of K⁺ channels is believed to cause depolarization of the presynaptic membrane and slow down or inhibit repolarization. Prolonged depolarization results in opening of slow voltage-gated calcium (Ca²⁺) channels and allows a subsequent influx of Ca²⁺. The increased concentration of intracellular Ca²⁺ induces exocytosis of more synaptic vesicles containing acetylcholine (ACh), thus releasing an increased level of ACh into the synaptic cleft. The influx of ACh into the presynaptic cleft enhances neuromuscular transmission, providing improved muscle function.

Absorption: Amifampridine as phosphate is rapidly absorbed, and peak plasma concentration (C_{max}) is reached 20 minutes to 1 hour after ingestion. Food has an effect on pharmacokinetics of amifampridine. The extent of exposure (AUC) of amifampridine was decreased by approximately 18% to 20% and C_{max} was decreased by about 50% following administration with high fat meal as compared with administration in the fasted state. Amifampridine phosphate displays approximately dose-proportional pharmacokinetics following single oral doses of 20 mg to 80 mg. this drug can be administered without regard to food.

Distribution: The estimated apparent volume of distribution of amifampridine (V/F) was 467 L. Nonclinical in vitro data from studies with human plasma indicate a lack of protein binding for amifampridine.

Metabolism and Elimination: Amifampridine is metabolized by N-acetyltransferase to 3-N-acetyl-amifampridine, which is considered as an inactive metabolite. The rate and extent of metabolism is affected by polymorphism in the NAT2 gene. Slow acetylators have 3 to 5 times higher plasma amifampridine levels than fast acetylators. The terminal half-life ranges from 1.8 to 2.5 hours in healthy subjects for fast and slow acetylators, respectively. About 93% to 100% of the administered dose is eliminated in the urine as amifampridine or 3-N-acetyl amifampridine over 24 hours. Urinary excretion data in humans indicated that on average, 19% of the unchanged amifampridine was eliminated in the urine in 24 hours after administration of amifampridine phosphate. Furthermore, 74.0% to 81.7% of 3-N-acetyl amifampridine was eliminated in 24 hours after amifampridine phosphate administration.

Specific Populations:

Renal Impairment: The exposure of amifampridine was 2-3 fold higher in subjects with moderate and severe renal impairment, compared to subjects with normal renal function. Subjects with mild renal impairment had 36% increase in exposure compared with subjects with normal renal function. Dose titration from a low starting dose of 15 mg is recommended for subjects with impaired renal function.

Geriatric: In clinical efficacy studies, 25.4% (16 of 63) of LEMS patients receiving FIRDAPSE® were ≥ 65 years of age. No major safety concerns arose from the clinical experience with amifampridine in this population.

Acetylation Status and Polymorphism: The rate and extent of metabolism of amifampridine is affected by polymorphism in the N-acetyltransferase gene 2 (*NAT2*). Slow acetylators have 3-5 times higher plasma drug levels than fast acetylators. In the general population, the *NAT2* slow acetylator phenotype is more common among European and North American natives, including Caucasians and African populations (40–70%) than those with Pacific Asian ethnic backgrounds (10–30% among Japanese, Chinese, Korean, and Thai).

The individualized dose titration, with a lower starting dose for subjects who are known slow acetylators, based on tolerability and effect should account for the individual differences in FIRDAPSE® metabolism.

Hepatic Impairment: Amifampridine is mainly eliminated through hepatic metabolism. However, the effect of hepatic impairment on the exposure of amifampridine was not studied.

The applicant has committed to conduct the hepatic impairment study post approval. Patients with ALT/AST >5 times the ULN and/or total bilirubin >3 times the ULN were excluded from clinical studies. Similar to dosing in known slow acetylators, patients with hepatic impairment are recommended to start dose titration with 15 mg per day.

2.2 Dosing and Therapeutic Individualization

2.2.1 General dosing

The recommended starting dose of FIRDAPSE® is 15- 30 mg a day taken orally in divided doses 3 to 4 times per day. The dose can be increased by 5 mg per day every 3 or 4 days. Dose should not exceed a maximum of 80 mg per day. The recommended maximum single unit dose is 20 mg. FIRDAPSE® can be taken without regard to food. If a dose is missed, patients should not take double or extra doses.

2.2.2 Therapeutic individualization

The applicant conducted phase I clinical pharmacology studies to evaluate the effect of intrinsic/extrinsic factors on the exposure of amifampridine. In addition, in-vitro studies were conducted to evaluate potential drug interactions with major enzymes and transporters for both parent and major inactive N-acetyl metabolite. Results from in-vitro studies suggested that amifampridine is not a substrate or an inhibitor/inducer of major CYP enzymes or transporters. Also, the major metabolite was not an inhibitor or inducer of major enzymes or transporters. Therefore, no clinical drug-drug interaction studies were conducted. The need for dose individualization based on the tested intrinsic factors is discussed below:

Renal Impairment: No dose capping is needed for FIRDAPSE® by renal function. The pharmacokinetics, safety, and tolerability of a single 10 mg dose of FIRDAPSE® was evaluated in patients with mild, moderate and severe renal impairment (Study REN001). The exposure of amifampridine, measured as AUC, was approximately 3-fold higher in subjects with severe renal impairment compared with subjects with normal renal function. The C_{max} of amifampridine increased less than 1.5-fold in subjects with renal impairment compared with subjects with normal renal function. The review team concluded that individual dose titration based on clinical effect and tolerability should account for these effects. Moreover, the efficacy studies enrolled patient with mild and moderate renal impairment and they were titrated to a tolerated effective dose. Patients with renal impairment are recommended to start dose titration with 15 mg per day.

Hepatic Impairment: The effects of FIRDAPSE® have not been studied in patients with hepatic impairment and no specific dose adjustment recommendation can be made for patients with hepatic impairment. Hepatic impairment is expected to increase the plasma concentration of amifampridine, however, the recommended dose titration should facilitate individualization of dose for these patients. The applicant has committed to conduct a hepatic impairment study after the approval of FIRDAPSE®. Patients with hepatic impairment are recommended to start dose titration with 15 mg per day.

N-acetyltransferase 2 (NAT2) Phenotype: The rate and extent of metabolism of amifampridine is affected by polymorphism in the N-acetyltransferase gene 2 (NAT2). Slow acetylators have 3-5 times higher plasma drug levels than fast acetylators. In the general population, the NAT2 slow acetylator phenotype is more common among European and North American natives, including Caucasians and African populations (40–70%) than those with Pacific Asian ethnic backgrounds (10–30% among Japanese, Chinese, Korean, and Thai). The individualized dose titration from a 15 mg dose to a tolerated and effective dose in subjects with known slow acetylator status should account for individual differences in FIRDAPSE® metabolism.

Geriatric Population: In clinical efficacy studies, 25.4% (16 of 63) of LEMS patients receiving FIRDAPSE® were ≥ 65 years of age. No special safety concerns arose from the clinical experience with amifampridine in this population. In a population PK analysis, including healthy subjects and patients, with ages ranging from 18 to 80 years, patient age was not found to influence the oral clearance of amifampridine.

2.3 Outstanding Issues

None

2.4 Summary of Labeling Recommendations

- The recommended starting dose of FIRDAPSE® should be 15 - 30 mg a day (b) (4), taken orally in divided doses 3 to 4 times per day. The dose can be increased by 5 mg per day every 3 or 4 days. The low starting dose should be considered for patients who are known slow acetylators or for patients with renal/hepatic function impairment.
- The maximum unit dose allowed is 20 mg.
- There is no daily dose capping recommended for patients with renal impairment or for patients who are slow acetylators.
- The drug can be administered without regard to food.

3. COMPREHENSIVE CLINICAL PHARMACOLOGY REVIEW

3.1 Overview of the Product and Regulatory Background

FIRDAPSE® was approved for the treatment of LEMS by the European Medicines Authority (EMA) and has been marketed in the European Union by BioMarin Pharmaceutical, Inc. (BioMarin) since December 2009. BioMarin transferred the sponsorship of amifampridine phosphate to Catalyst in March 2013. In the United States, amifampridine has been available through compounding pharmacies for the treatment of LEMS. There are two forms available of amifampridine. The first one is the free base amifampridine and the second is the phosphate salt (FIRDAPSE®). There are other names used for amifampridine: 3,4-DAP phosphate, 3,4-pyridinediamine, phosphate (1:1); pyridine-3,4-diamine phosphate; diamino-3,4-pyridine, phosphate salt; and 3,4-diaminopyridine phosphate. The studies that evaluated the clinical pharmacology and efficacy of amifampridine are summarized in Table 1.

Table 1: Summary of Individual Clinical Pharmacology and Efficacy Studies

Study	Study Type	Subject Population	Dosage and Route of Administration of Amifampridine	Treatment Duration
Legacy Studies				
DAPSEL	Phase 1	Healthy	20 mg base equivalent of free base or phosphate salt, oral	Single dose
Sponsored Studies				
LMS-001	Phase 1	Healthy	20 mg, 2 oral doses: 1 dose fasted and 1 dose fed	2 d
FIR-001	Phase 1	Healthy (fast and slow acetylators)	Part 1: ascending oral doses of 5, 10, 20, 30 mg Part 2 Group 1: oral dose of 20 mg QID Part 2 Group 2: oral dose of 20 mg QID	Part 1: 4 d Part 2, Group 1: 1 d Part 2, Group 2: 4 d
QTC-002	Phase 1	Healthy	Part 1: ascending weekly oral doses of 30, 60, 80, and 70 mg Part 2: 2 oral doses 30 mg, 2 oral doses 60 mg, administered every 4 days	Part 1: 4 w Part 2: 13 d
REN-002	Phase 1	Subjects with mild, moderate, severe RI or normal renal function	Single oral dose of 10 mg	1 d
LMS-002	Phase 3	LEMS	30-80 mg/d given TID or QID, maximum of 20 mg per dose, oral	Part 1: 7-91 d Part 2: 7 d Part 3: 7 d Part 4: open-label, long-term safety
LMS-003	Phase 3	LEMS	30-80 mg/d given TID or QID, maximum of 20 mg per dose, oral	4 days

3.2 General Pharmacology and Pharmacokinetic Characteristics

Pharmacology	
Mechanism of Action	Amifampridine is a voltage-gated potassium (K^+) channel blocker. The blockade of K^+ channels causes depolarization of the presynaptic membrane and slows down or inhibits repolarization. Prolonged depolarization results in opening of slow voltage-gated calcium (Ca^{2+}) channels and allows a subsequent influx of Ca^{2+} . The increased concentration of intracellular Ca^{2+} induces exocytosis of more synaptic vesicles containing acetylcholine (ACh), thus releasing an increased level of ACh into the synaptic cleft. The influx of ACh into the presynaptic cleft enhances neuromuscular transmission, providing improved muscle function.
Active Moieties	There are no known active metabolites for amifampridine. There was one major inactive metabolite reported (3-N-acetyl amifampridine). The mean C_{max} and AUC values for 3-N-acetyl amifampridine were approximately 4.5-fold and 12-fold higher, respectively, than the C_{max} and AUC values for amifampridine. The mean plasma $T_{1/2}$ for 3-N-acetyl amifampridine was longer than that for amifampridine (4.0 hours for 3-N-acetyl amifampridine vs. 1.8 hours for amifampridine).
QT Prolongation	The effect of a therapeutic dose and a relative supra-therapeutic dose of FIRDAPSE® on QTc interval prolongation was studied in a double-blind, randomized, placebo and positive controlled study in 60 healthy subjects. FIRDAPSE® was not associated with any clinically meaningful effect on heart rate, AV conduction, cardiac repolarization, or QRS interval duration at individual doses ranging from 30 mg to 60 mg. However, there is a theoretical concern that concomitant use of FIRDAPSE® and drugs that cause QT prolongation may lead to an increased risk of arrhythmia.
General Information	
Bioanalysis	Amifampridine(3,4-diaminopyridine) was analyzed using a validated LCMSMS method. Please refer to Section 4.1 for a summary of bioanalytical method validation and its performance.
Dose Proportionality	Amifampridine displays approximately dose-proportional pharmacokinetics following single oral doses of 20 mg to 80 mg.

Accumulation	There was no evidence of accumulation after multiple dosing. This is expected given the short elimination half-life (T _{1/2}) for amifampridine.
Variability	There was significant inter-subject variability in amifampridine levels. The coefficients of variation for C _{max} and AUC were approximately 60% to 70%.
Bioavailability	The absolute bioavailability of amifampridine was not reported. However, 93% to 100% of the administered dose is recovered in the urine over 24 hours as parent or metabolites
T_{max}	Amifampridine was rapidly absorbed at both the 10- and 20-mg dose levels, with T _{max} ranging from 0.5 to 2 hours after dosing.
Food effect (high-fat meal) GMR relative to fasted state for tablets	Administration of amifampridine phosphate with food resulted in reduction in the mean exposure of amifampridine; mean C _{max} was reduced by approximately 40% and mean AUC was reduced by 20%.
Volume of Distribution	The estimated apparent volume of distribution of amifampridine (V/F) was 467 L.
Plasma Protein Binding	In vitro data from studies with human plasma indicate the lack of protein binding for amifampridine.
Substrate transporter systems	Amifampridine and its major metabolites are not substrates or inhibitors for major drug transporters based on in-vitro studies.
Elimination	
Mean Terminal Elimination half-life	The average plasma elimination half-life is ranges from 1.8-2.5 hours for amifampridine and 4 hours for the 3-Nacetylated amifampridine metabolite.
Metabolism	
Primary metabolic pathway(s) [<i>in vitro</i>]	In vitro and in vivo studies in humans indicate that amifampridine is metabolized to a single major 3- N-acetylated amifampridine metabolite through N Acetyl transferase (NAT-2) enzyme.

Inhibitor/Inducer	In-vitro data showed that amifampridine and 3-N-acetylated amifampridine metabolite are not inhibitors or inducers of major CYP enzyme.
Excretion	
Primary excretion pathways	In humans, 93.2% to 100% of amifampridine is excreted into the urine within 24 hours after dosing as amifampridine (19%) and its 3-N-acetylated amifampridine metabolite (74.0% to 81.7%). The plasma elimination half-life is approximately 1.8-2.5 hours for the amifampridine and 4 hours for the 3-N-acetylated amifampridine metabolite.

3.3 Clinical Pharmacology Review Questions

3.3.1 To what extent does the available clinical pharmacology information provide pivotal or supportive evidence of effectiveness?

The primary evidence of effectiveness of amifampridine phosphate for the treatment of LEMS is based on results from the following Phase III studies:

Study LMS-002: This was a randomized, double-blind, placebo-controlled withdrawal design, pivotal Phase 3 study (N ~38). This study consisted of 4 parts. Part 1 was an open-label, run-in phase at the beginning of the study and was designed to allow subjects to be titrated to achieve an optimal dose with a starting daily dose of 15 mg/day (5 mg TID) up to 80 mg total daily dose given four times daily. After patients were stabilized on an optimal dose in part 1, they were randomized to the discontinuation phase (Part 2). In Part 2 of Study LMS-002, subjects were randomly assigned to blinded treatment that consisted of either continuation of amifampridine phosphate or downward titration to 0 mg (placebo) over a 7-day period. Subjects continued blinded treatment for an additional 7 days in Part 3 for a total of 14 days in the randomized, blinded and placebo controlled portion of the study before they were transferred to part 4 of the study (open label extension). Primary and secondary endpoints were calculated based on change from baseline over 14 days from the start of the discontinuation phase (Part2) until the end of the double-blind treatment phase (Part 3).

Study LMS-003: This was a supportive Phase 3, randomized, double-blind, placebo-controlled withdrawal study, which enrolled 26 male and female patients across 3 study sites in the United States. Immediately before participation in the study, patients were receiving open-label treatment with amifampridine phosphate in the expanded-access program (Study EAP-001). Subjects were required to be on a stable optimal dose and dosing frequency of amifampridine phosphate for at least 1 week before being randomized (Day 0) to either placebo or

amifampridine phosphate. Then patients were treated for four days before assessment was made for primary and secondary endpoints as explained below.

In Study LMS-002 and Study LMS-003, efficacy was assessed by evaluating the changes in muscle strength and LEMS symptoms, and by evaluating the changes in global impression scales from investigators and from subjects. In both studies, the co-primary efficacy endpoints were change in the QMG score and the change in the SGI score. The results from the two-pivotal randomized, controlled clinical trials that evaluated amifampridine phosphate in LEMS (Study LMS-002 and Study LMS-003) are summarized in Table 2 & Table 3.

Table 2: Summary of Change from Baseline in Primary and Secondary Efficacy Endpoints for Study LMS-002 from Day 1 (Part 2) to Day 14 (End of Part 3) in the ITT population

Assessment	Amifampridine phosphate	Placebo
	N = 16	N = 21
Quantitative Myasthenia Gravis Scores		
LS mean change from baseline	0.4	2.2
Difference in LS means (amifampridine phosphate minus placebo (95% CI)	−1.7 (−3.4, −0.0)	
p-value	0.0452	
Subject Global Impression Scores		
LS mean change from baseline	−0.8	−2.6
Difference in LS means (amifampridine phosphate minus placebo (95% CI)	1.8 (0.7, 3.0)	
p-value	0.0028	

Source: Table 11.2/11.3 ; Page Page 91 & 93 of 2369; LMS-002 Clinical Study Report

Table 3: Summary of Change from Baseline in Primary and Secondary Efficacy Endpoints for Study LMS-003 from Day 0 to Day 4 (ITT population)

Assessment	Amifampridine phosphate	Placebo
	N = 13	N = 13
Subject Global Impression Scores		
LS mean change from baseline	−0.64	−.59
LS means difference (95% CI)	2.95 (1.53, 4.38)	
p-value	0.0003	
Quantitative Myasthenia Gravis Scores		
LS mean change from baseline	0.00	6.54

LS means difference (95% CI)	-6.54 (-9.78, -3.29)
p-value	0.0004

Source: Table 11.4/11.22 ; Page Page 43 & 93 of 182; LMS-002 Clinical Study Report;

In addition to the above two studies, the sponsor submitted literature reports for several clinical studies and some case reports from the past 25 years describing the use of amifampridine for the treatment of LEMS. This included five published, randomized, double-blind, placebo-controlled trials; one randomized, active-controlled trial; a meta-analysis of four controlled clinical trials; and data from uncontrolled studies and case reports.

3.3.2 Is the proposed dosing regimen appropriate for the general patient population for which the indication is being sought?

The proposed dosing regimen is generally acceptable except for the (b) (4) dose as the only starting dose. The proposed dosing regimen for FIRDAPSE® is based on a titration approach from a starting total daily dose of (b) (4) to an optimal dose not exceeding 80 mg/day given in 3 to 4 divided doses, with no single dose of more than 20 mg, and titrated to neuromuscular benefit as determined by the clinician.

The single maximum dose of 20 mg was chosen because it is same as the maximum single dose administered in the efficacy trials (LMS-002 and LMS003). Additionally, several randomized, placebo controlled studies in patients with LEMS have demonstrated the safety and tolerability of a maximum single dose of 20 mg amifampridine¹. Due to the short mean half-life (t_{1/2}=1.8h) and short duration of action, the total daily dose is divided so that it is taken 3 or 4 times a day.

The proposed lowest total daily dose of amifampridine phosphate is (b) (4) per day (divided doses, maximum 20 mg per single dose). The review team recommended changing the starting total daily dose from (b) (4) to a dose range of 15-30 mg/day dose. This is particularly important for patients with intrinsic factors that are expected to increase the amifampridine plasma concentration. This recommendation was based on the following reasons:

Giving a lower starting dose for patients with intrinsic factors (e.g. known slow acetylators status, renal/hepatic impairment) will ensure gradual increase to an optimal dose. This is important because there is no specified dose adjustments for patients based on intrinsic factors like renal impairment or acetylation status. In addition, patients with CrCL<49 ml/min were not tested in the submitted clinical trials and the dedicated renal impairment study showed about 2-3 fold increase in exposure for patients with moderate and severe renal impairment. Also, the starting dose specified in the protocol for study LMS-002 was 15 mg/day and not (b) (4). Therefore, we recommend a lower starting dose for those subjects with renal/hepatic impairment or who are known slow acetylators.

The safety of amifampridine at the maximum dose of 80 mg per day is supported by numerous published controlled and uncontrolled studies and case reports over the last 25 years. In

¹ Jaretzki A 3rd, Barohn RJ, Ernstoff RM, Kaminski HJ, Keeseey JC, Penn AS, Sanders DB. Neurology. 2000 Jul 12;55(1):16-23.

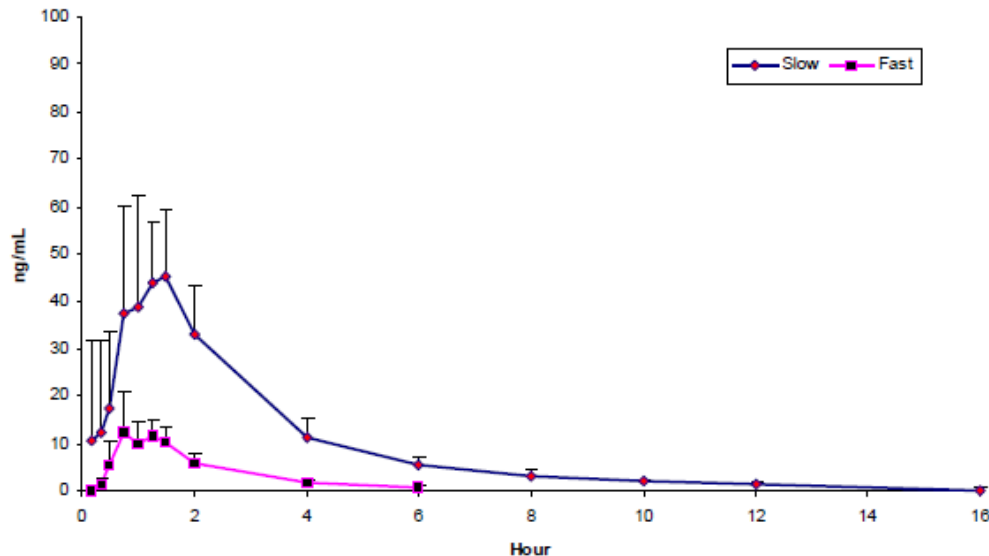
addition, 4 patients who were known slow acetylators were stabilized at the 80 mg total daily dose with no significant adverse events reported in clinical trials. Thus, capping the highest dose for slow acetylators or renally impaired patients is not required. It should be noted that the aggregated clinical experience suggests that there may be an increased risk of seizure with amifampridine, no such incidence were reported in clinical trials. Please refer to the clinical safety review for additional details (Dr.Veneeta Danton clinical review).

3.3.3 Is an alternative dosing regimen and/or management strategy required for subpopulations based on intrinsic factors?

No specific dose adjustment is recommended for patients based on intrinsic factors. However, the review team recommends initiating the dose titration with 15 mg dose for subjects with renal/hepatic impairment or who are known slow acetylators. Individual dose titration based on tolerability and efficacy should account for the individual differences in amifampridine exposure. The effect of intrinsic factors on the exposure of amifampridine is explained below.

Acetylation Status: The pharmacokinetics and systemic exposure to amifampridine is notably influenced by the overall metabolic acetylation activity of the NAT2 enzyme. The NAT2 gene is highly polymorphic and results in phenotypes with variable acetylation activity rates ranging from slow to fast. In the healthy volunteer study (i.e., study FRI-001) fast acetylators were defined by having a caffeine metabolite ratio >0.3 and slow acetylators with a caffeine metabolite ratio <0.2 . This was a single dose (20 mg) and a multiple dose study of FIRDAPSE® for up to 3 days. In the MAD part of the study amifampridine was administered once daily for three days. Subjects who are slow acetylators had up to 3-5 fold higher exposure compared to fast acetylators (Figure 1). No dose adjustment or dose capping is recommended based on acetylation status because it is expected that individual dose titration based on tolerability and efficacy will address this issue. In addition, there were 4 out of 16 patients in the treatment arm who are known slow acetylators and were taking up to 80 mg total daily dose to achieve optimal response. The applicant's classification of phenotypes assignments based on the caffeine challenge test and utilized genotype methodologies are generally acceptable. Of note, while there are no FDA-approved or cleared NAT2 genotyping assays, NAT2 status testing is available in certain CLIA certified laboratories. In the pivotal efficacy study (LMS002) 9 out of 16 patients were slow acetylators and were stabilized at different steady state doses ranging from 30-80 mg total daily dose. No specific safety event was noticed in the efficacy study LMS002 even for the 4 slow acetylator patients stabilized at the highest dose (80 mg/day). More safety events were observed in Phase I study FIR001. This is mainly because the dose was the same for all the volunteers (20 mg single or once daily dose for 3 days) and was not titrated based on tolerability. In conclusion, no warning or dose adjustment recommendation is needed for patients based on acetylation status. Patients who are slow acetylators are recommended to start dose titration with a 15 mg total daily dose.

Figure 1: Mean Plasma Concentration-Time Profiles of Amifampridine After Single Oral Dose of 20 mg (base equivalents) in Healthy Subjects with Slow and Fast Acetylator Phenotypes.



Source: Figure 6., Page1426 of 2062 study-report-FIR-001.

Renal Impairment: Exposure of amifampridine was generally higher in subjects with renal impairment than in subjects with normal renal function; however, NAT2 phenotype had a greater effect on an individual's exposure to amifampridine than renal function status (Table 4). Amifampridine exposure by $AUC_{0-\infty}$ was up to 2-fold higher in slow acetylators and up to 3-fold higher in fast acetylators with severe renal impairment compared to subjects with normal renal function. Exposure by C_{max} was marginally affected by renal impairment regardless of acetylation status. No dose adjustment is recommended based on renal impairment status because individual dose titration is expected to provide optimal dose for each patient. The increase in exposure observed for patients with severe renal impairment is less than the increase seen in patients who are slow acetylators with normal renal function, as mentioned in the above paragraph.

Table 4. Summary of Amifampridine PK in Subjects with Renal Impairment and Normal Renal Function from Study REN-002

Renal Function Group					
		Mild	Moderate	Severe	Normal
AUC _{0-∞} (h·ng/mL)					
n	SA	4	4	4	4
	RA	3 ^a	3 ^a	2 ^a	3 ^a
Mean (SD)	SA	81.29 (33.371)	126.06 (17.510)	118.60 (37.119)	59.07 (10.278)
	RA	16.05 (3.461)	14.34 (6.801)	32.76 (29.468)	10.73 (0.204)
C _{max} (ng/mL)					
n	SA	4	4	4	4
	RA	4	4	4	4
Mean (SD)	SA	33.48 (13.104)	52.53 (5.163)	44.05 (12.877)	38.63 (9.155)
	RA	11.08 (4.686)	8.33 (2.739)	9.48 (5.299)	7.65 (3.233)
CL/F (L/h)					
n	SA	4	4	4	4
	RA	4	4	3	4
Mean (SD)	SA	137.3 (48.5)	80.4 (10.7)	90.9 (28.8)	173.8 (34.7)
	RA	667.4 (133.0)	864.6 (282.5)	585.7 (349.7)	1077.3 (290.4)

SA = slow acetylation status; SD = standard deviation; Source: [Study REN-002, Table 14.2.2.1](#).

Number of subjects <4 in cases where the terminal phase of individual PK was not suitable for calculation of the parameter. Source: Table 6; Page 57 of 544 Clinical Study Report REN-002/CPX500PC-135006

Although no specific dose adjustment will be recommended, these patients should be started at a lower starting dose (e.g. 15 mg/day). Also, close monitoring should be made while taking the maximum dose for patients who are both renally impaired and are known slow acetylators. In the clinical trial LMS002 there were only 2 patients with moderate renal impairment who were slow acetylators and they were stabilized at 40 and 50 mg total daily dose.

Hepatic Impairment: Although amifampridine is eliminated through hepatic metabolism, the effect of hepatic impairment on the exposure of amifampridine was not studied. In addition, patients with ALT/AST >5 times the ULN and/or total bilirubin >3 times the ULN were excluded from clinical studies. No specific dose adjustments for patients with hepatic impairment can be made and they should be closely monitored while dosing. Similar to patients who are known slow acetylators, patients with hepatic impairment are recommended to start dose titration slowly with 15 mg per day.

3.3.4 Are there clinically relevant food-drug or drug-drug interactions and what is the appropriate management strategy?

In humans, the rate and extent of absorption of amifampridine is influenced by food. There was a decrease in C_{max} and AUC, and an increase in the time to reach maximum plasma concentrations when amifampridine phosphate was administered with food as compared to without food.

Overall, food slowed and decreased the absorption of amifampridine with a lowering of C_{max} on average by ~44% and AUC by ~20%. Additionally, mean T_{max} was shorter in the fasted state (0.6 hours) compared with the fed state (1.3 hours), indicating that food slows and decreases absorption of amifampridine phosphate. (b) (4)

, the review team recommends that FIRDAPSE® should be taken without regard to food based on the following:

The pivotal efficacy study (LMS002) FIRDAPSE® was given with food while in the second study (LMS003) FIRDAPSE® was administered without regard to food. Both studies showed positive safety and efficacy results. In addition, food results in a decrease in AUC by only 20% and C_{max} by about 44%. Moreover, we believe it is less practical in real world to administer FIRDAPSE® tablets with food multiple times a day. Therefore, FIRDAPSE® tablets can be taken without regard to food.

4. APPENDICES

4.1 Summary of Bioanalytical Method Validation and Performance

Plasma concentrations of amifampridine was determined using protein precipitation followed by high-performance liquid chromatography (HPLC) and tandem mass spectrometric detection

(b) (4) Briefly, 25 uL of plasma sample was mixed with 300 uL of ACN containing the internal standards and 0.1% formic acid. After centrifugation, 200 uL of supernatant was transferred to a clean 96-well plate and injected into the LC/MS/MS system. The method employed individual stable isotopelabeled internal standards for amifampridine ([²H₃]-3, 4- diaminopyridine). The details of the bio-analytical methods used in this NDA are presented in Table 5. The methods satisfied the criteria for method validation and application to routine analysis set by the Guidance for Industry: Bioanalytical Method Validation, and hence are acceptable.

Table 5: Bioanalytical Assay Performance Characteristics

Analyte/Parameter	Amifampridine
Range (ng/ml)	0.500 – 500 ng/mL
Inter Batch Precision (% CV)	7.6%
Inter Batch Accuracy (%)	99.4%
Internal standard (IS)	[² H ₃]-3, 4-diaminopyridin
Reference standard	Amifampridine
Selectivity	No significant interference for reference or the internal standard in any of the human plasma (EDTA) lots screened
Recovery (%)	87.7% – 94.1%
<u>Stability (%)</u>	
Freeze/Thaw Stability	5 Cycles (-10 to -30°C) (83-86%)
At room temperature (% difference)	24 hrs 2.8%-1.8%
Stock solution 2-8 C	1.8%
% Difference for amifampridine:	

4.2 Pharmacogenomic Report

According to the applicant, amifampridine is metabolized primarily by acetylation, which is highly variable due to N-acetyltransferase 1 (NAT1) and N-acetyltransferase 2 (NAT2) genetic polymorphisms. Amifampridine exposure and C_{max} values are 3- to 5-fold greater in slow acetylators compared to rapid acetylators in both healthy subjects and patients diagnosed with LEMS.

NAT2 polymorphisms are known to result in variable acetylation activity (i.e., slow vs. intermediate vs. fast acetylators) within and across different racial and ethnic populations. Published frequencies for NAT2 phenotypes are listed below (Table 1). Frequencies reported in different studies may vary depending on the performance characteristics of the NAT2 genotyping assay used, on which alleles were genotyped, and on how the genotypes were grouped into phenotype categories.

Table 1. Frequency of NAT2 phenotypes across racial/ethnic populations

Racial/Ethnic Population	NAT2 Phenotype Frequency (%)		
	Fast	Intermediate	Slow
Caucasian	6-7	35-40	56-59
African American	14-19	44-45	37-42
U.S. Hispanic	14	55	33
Japanese	44-45	46-49	7-10
U.S. Korean	46	40	13
Chinese	23-27	48-53	21-27

Source: Modified based on PMID: 20183529. Fast acetylator (2 fast alleles)= homozygous or heterozygous for NAT2*4, *12A, *13A; Slow acetylator (2 slow alleles)= homozygous or heterozygous for NAT2*5-*7, *14; Intermediate acetylator (1 fast and 1 slow alleles).

The purpose of this review is to determine if the NAT 1 and NAT2 phenotype assignments as performed by the applicant are acceptable.

CONTENTS

The overall genotype data for the NAT2 gene performed in subjects from 4 clinical trials are shown in Table 2.

Table 2. Listing of clinical studies with available NAT2 acetylator status information

Study Identifier	Number of Subjects (n)	NAT2 Acetylator Status (n)		
		Fast	Intermediate	Slow
FRI-001	26	13	N/A	13
QTC-002	60	N/A	N/A	60
REN-002	32	16	N/A	16
LMS-002	38*	3	9	25

*One patient didn't consent for genetic testing. N/A - not applicable.

CLINICAL STUDIES

Study FRI-001 included pharmacogenomic analyses of healthy subjects administered amifampridine phosphate. According to the applicant, slow acetylators had 3.5-fold higher mean amifampridine C_{max} and 5.5-fold higher mean amifampridine AUC compared to fast acetylators after a single dose of amifampridine phosphate; plasma half-life was approximately 2-fold longer in slow acetylators compared to fast acetylators in healthy subjects.

Methods

Phenotyping test:

A caffeine challenge test was performed via collection of urine samples for 6 hours following a single oral dose of 150 mg caffeine on Day -7. 5-acetylamino-6-formylamino-3-methyl uracil (AFMU), acetylamino-6-amino-3-methyluracil (AAMU), 1-methylxanthine (1X) and 1-methylurate (1U) were determined in urine by liquid chromatography-mass spectrometry/mass spectrometry (LC MS/MS).

All subjects were divided into 2 groups based on N-acetylated caffeine metabolite ratio:

- slow (ratio <0.2)
- fast (ratio >0.3)

Subjects with N-acetylated caffeine metabolite ratios of ≥ 0.2 and ≤ 0.3 (intermediate acetylators) were excluded from the FRI-001 study.

Genotyping test:

The NAT2 genotypes were determined using the StepOnePlus Real Time PCR system from Applied Biosystems (TaqMan single-nucleotide polymorphism [SNP] genotyping with allelic discrimination).

All subjects were screened for 2 mutations (282C>T and 341T>C) in the NAT2 gene.

Subjects were assigned into 2 classes of acetylator activity:

- 1 or 2 fast alleles (fast acetylator)
- no fast alleles (slow acetylator)

Table 3 provides possible NAT2 phenotype status assignments. Patient level genotype status was provided by the applicant.

Table 3. A summary of the possible NAT2 gene polymorphisms and assigned phenotype status

T341C	C282T	Assigned Phenotype Status
Wildtype	Wildtype	Fast
Wildtype	Heterozygote	Fast
Wildtype	Mutant	Slow
Heterozygote	Wildtype	Fast
Heterozygote	Heterozygote	Slow
Heterozygote	Mutant	Slow
Mutant	Wildtype	Slow
Mutant	Heterozygote	Slow
Mutant	Mutant	Slow

Source: FRI-001 Study Report, Table, Page 421.

Table 4 provides a summary of the acetylase assignment based on the caffeine challenge and genotype results for 282C>T and 341T>C variants.

Table 4. A summary of NAT2 genotype and phenotype status

		Genotype: Mutant C282T/T341C							
		Acetylase Type		Fast			Slow		
Part	Group	Fast	Slow	WT/WT	HE/WT	WT/HE	HE/HE	MU/WT	WT/MU
1		6 (50%)	6 (50%)		3 (25%)	3 (25%)	1 (8%)	3 (25%)	2 (17%)
2	1	2 (50%)	2 (50%)		2 (50%)		2 (50%)		
	2	5 (50%)	5 (50%)	2 (20%)	1 (10%)	2 (20%)	3 (30%)	2 (20%)	

WT= wildtype; HE= heterozygote; MU= mutant

Source: FRI-001 Study Report, Table 10.4, Page 69.

Reviewer's comments:

The applicant's classification of phenotype based on genotype seems reasonable based on the Arylamine N-acetyltransferases (NATs) database (<http://nat.mbg.duth.gr/>). Also, according to the applicant, the metabolic acetylation phenotypes (fast or slow acetylase status) based on the caffeine challenge test, matched with the phenotype status assigned based on the NAT2 genotyping results for all subjects. (Table 4).

However, there are several limitations regarding the database used and the variants genotyped by the applicant that should be noted: 1) the applicant (only) genotyped the 341T>C variant and the 282C>T variant, which do not cover all existing NAT2 haplotypes; a correct haplotype, i.e., "*" allele designation, may be limited given that both variants exist in multiple haplotypes; 2) certain NAT2 haplotypes may not have an assigned phenotype given the limited data available in the literature; 3) in some cases, phenotype assignment may be substrate dependent (e.g., *7B, slow acetylase).

Study QTC-002 enrolled only subjects who were slow acetylators.

Methods

Genotyping test:

Description of the genotype methodology was not provided in the study protocol or study report.

Reviewer's comments:

According to the applicant, study QTC-002 enrolled only subjects with slow acetylator status. Genotype testing methods were not described and genotype results were not submitted. The reviewer was unable to verify phenotype results.

Study REN-002 the effects of renal impairment on the PK and tolerance of amifampridine were evaluated and compared with subjects with normal renal function. Additionally, subjects were stratified by NAT2 acetylator status. Both, AUC_{0-inf} and C_{max} were 5-fold higher in slow acetylators compared to rapid acetylators.

Study REN-002 enrolled 4 slow and 4 fast acetylators within each renal function group (mild, moderate, severe, and normal renal function). Acetylator status were determined by a caffeine challenge and NAT2 genotyping.

Methods

Phenotyping test:

Performed by examination of caffeine metabolites AFMU/1X by high-performance liquid chromatography (HPLC) or LCMS/MS. No information about phenotype assignments based on the caffeine challenge was provided in the study protocol or study report.

Genotyping test:

Performed using a combination of direct DNA sequencing or quantitative polymerase chain reaction (qPCR) (Taqman) depending upon the mutation. All subjects were genotyped for NAT2 in order to access the acetylation status.

Reviewer's comments:

The applicant did not provide patient level NAT2 genotype and/or phenotype status. Therefore, the reviewer was unable to verify acetylator status assignments.

Study LMS-002 the efficacy, safety, and PK of amifampridine administered as the phosphate salt in subjects with LEMS were determined. Findings were consistent with results in healthy subjects, mean AUC₀₋₄ values calculated for slow acetylators were approximately 2-fold and 4-fold higher than the mean AUC₀₋₄ for intermediate and fast acetylators, respectively; mean C_{max} values calculated for slow acetylators were approximately 2-fold and 5-fold higher than the mean C_{max} values for intermediate and fast acetylators, respectively.

Methods

Genotyping test:

The genotyping procedures were carried out by (b) (4). Sanger sequencing was used to genotype the NAT1 and NAT2 genes. Of patients who completed the study (N=38), 37 patients consented to the NAT1 and NAT2 analysis.

Tables 5 and 6 provide list of assessed NAT1 and NAT2 mutations (respectively) and assigned phenotype status.

Table 5. NAT1 assessed mutations and phenotype assignment

Gene	Star	Phenotype	Mutations
NAT1	*4	normal	given when no requested mutations are found
NAT1	*3	normal	1095 C>A
NAT1	*10	fast	1088 T>A and 1095 C>A
NAT1	*14A	slow	1088 T>A and 1095 C>A and 560 G>A
NAT1	*14B	slow	560 G>A
NAT1	*15	slow	559 C>T
NAT1	*17	slow	190 C>T
NAT1	*11A	fast	445 G>A and 459 G>A and 640 T>G and 1095 C>A
NAT1	*11B	fast	445 G>A and 459 G>A and 640 T>G
NAT1	*11C	fast	459 G>A and 640 T>G and 1095 C>A

Fast Acetylator (F): If *10 or *11 alleles are found in the sample
 Slow Acetylator (S): If *14, *15 or *17 alleles are found in the sample.
 Normal Acetylator (N): If only *4 or *3 alleles are found in the sample.

NB: Samples with *10/*14 genotype are considered as normal acetylators (N)

Source: (b) (4) Genotype Report, Table 7.2, Page 7.

Table 6. NAT2 assessed mutations and phenotype assignment

Gene	Star	Phenotype	Mutations
NAT2	*4	Fast	-
NAT2	*5D	Slow	341 T>C
NAT2	*6B	Slow	590 G>A
NAT2	*7A	Slow	857 G>A
NAT2	*14A	Slow	191 G>A

Fast Acetylator (F): two alleles "Fast"
 Intermediate Acetylator (I): one allele "Fast" and one allele "Slow"
 Slow Acetylator (S): two alleles "Slow"

Source: (b) (4) Genotype Report, Table 7.2, Page 7.

Tables 7 and 8 provide summary of the acetylator assignment based on the genotype results for the NAT1 and NAT2 genes, respectively.

Table 7. NAT1 genotypes and predicted phenotype status

Genotype	Predicted Phenotype	TOTAL
*10/*10	Fast	3
*4/*10	Fast	8
*3/*4	Normal	2
*4/*4	Normal	21
*4/*14A	Slow	1
*4/*15	Slow	2
TOTAL		37

(b) (6) did not consent to genetic testing for a total of N=37.
Fast Acetylator: If *10 or *11 alleles were found in the sample.
Slow Acetylator: If *14, *15 or *17 alleles were found in the sample.
Normal Acetylator: If only *4 or *3 alleles were found in the sample. Samples with *10/*14 genotype were considered as normal acetylators.

Source: LMS-002 Study Report, Table 11-18, Page 117.

Table 8. NAT2 genotypes and predicted phenotype status

Genotype	Predicted Phenotype	Total
*4/*4	Fast	3
*4/*5D	Intermediate	4
*4/*6B	Intermediate	5
*5D/*5D	Slow	8
*5D/*6B	Slow	11
*5D/*7A	Slow	1
*6B/*6B	Slow	4
*6B/*7A	Slow	1
TOTAL		37

(b) (6) did not consent to genetic testing for a total of N=37.
Fast Acetylator: two alleles "Fast".
Intermediate Acetylator: one allele "Fast" and one allele "Slow".
Slow Acetylator: two alleles "Slow".

Source: LMS-002 Study Report, Table 11-19, Page 117.

Reviewer's comments:

NAT1 genotype-phenotype assignment: According to the NATs database, haplotypes *3, *10, *11A, *11B, and *11C do not have assigned phenotype status based on limited evidence in available literature. Haplotypes *14A, *14B, *17 are categorized as slow acetylators (have lower activity than *4). Haplotype *15 is categorized as an allele with no enzyme activity based on the introduction of a stop codon causing truncation of the protein. The reviewer has limited ability to determine if the NAT1 phenotype assignments were performed correctly, specifically for the *3, *10, *11A, *11B, and *11C haplotypes, since variants listed in Table 5 can be present in multiple haplotypes.

According to the applicant, the enzymatic activity of the NAT1 may only play a secondary role in amifampridine metabolism due to an 8-fold preferential metabolic rate for the NAT2 over NAT1 enzyme.

NAT2 genotype-phenotype assignment: *The applicant's phenotype classification based on genotype seems reasonable based on the Arylamine N-acetyltransferases (NATs) database (<http://nat.mbg.duth.gr/>). Also, the variant selection seems adequate to cover the most common and relevant haplotypes of the NAT2 gene to adequately determine NAT2 phenotype. According to the NATs database, classification of subjects as slow acetylators by genotyping the 341T>C variant (i.e., *5D haplotype), the 590G>A variant (i.e., *6B haplotype), the 857G>A variant (i.e., *7A haplotype), or the 191G>A variant (i.e., *14A haplotype, not observed in the LMS-002 study) seems reasonable. Of note, per the NATs database, carriers of the 857G>A variant (i.e., *7A haplotype) should be assigned slow acetylator status; however, slow acetylator status based on *7A haplotype may be substrate dependent. Of note, the variants listed in Table 6 are present in multiple haplotypes. Some of the NAT database listed haplotypes do not have assigned phenotype status based on limited data in available literature.*

SUMMARY OF FINDINGS

Results obtained in healthy subjects and patients diagnosed with LEMS are consistent and showed that mean AUC₀₋₄ and C_{max} values are affected by the NAT2 acetylator status. The approach utilized by the applicant for determination of the acetylator status based on the caffeine phenotype assignment is acceptable. In addition, the NAT1 and NAT2 genotype methodologies, utilized by the applicant are generally acceptable. However, in some instances genotype assignments could not be verified. For example, no information about phenotype assignments based on the genotype results were provided in the REN-002 and QTC-002 study protocols or study reports.

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/s/

BILAL S ABU ASAL
09/13/2018

KATARZYNA DROZDA
09/13/2018

CHRISTIAN GRIMSTEIN
09/13/2018

KEVIN M KRUDYS
09/14/2018

SREEDHARAN N SABARINATH
09/14/2018

RAMANA S UPPOOR
09/14/2018